

Marginal-price support of electric-bus fleets with full battery replenishment

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Abstract

Electric-bus charging combines continuous energy decisions with indivisible service assignments. We ask whether a physical fleet schedule minimizes its bill at marginal electricity prices associated with its load. A subgradient support criterion accommodates multiple dispatch prices. Exact, fully replenished two-service constructions have positive gaps between physical and convexified planning under both quadratic supply and an hourly two-generator model with binding ramps. Dual coordination optimizes the hull bound; a numerical replay distinguishes improving dual bounds from switching physical responses. Replication can make the absolute planning gap vanish while whole-operator regret remains positive along a subsequence, although regret per reserved pair vanishes. This separates aggregate convexity from participant incentives. For two depot variants of one public timetable, analytical energy bounds strengthen computational enclosures in two archived markets. Those intervals still include zero gap, and responses diagnose returned incumbents whose optimality remains open. Timetable-scale certification remains the computational limitation.

Keywords: electric-bus scheduling; convex-hull pricing; marginal tariffs; charging coordination; nonconvex scheduling.

1 Introduction

An electric-bus operator must cover mandatory passenger services while assigning vehicles and scheduling charging. A tariff can shift charging toward cheaper periods, but a fleet large enough to affect aggregate electricity demand also changes the marginal cost behind that tariff. Even with continuously adjustable charging, its service assignments remain indivisible. The resulting question is whether a physically realizable fleet schedule minimizes the operator’s bill when evaluated at the marginal prices associated with that schedule’s load.

We study this question using complete fleet plans. Each plan includes all services, vehicle movements, charging and operating cost. Convexifying their joint cost and load vectors provides a useful planning lower bound, but a point in that hull is generally not an executable daily dispatch. Charging a convex supply cost at the mean load also differs from taking the mean cost of randomized physical dispatches. Maintaining these distinctions is essential both for interpreting a pricing failure and for assessing computational shortcuts.

The contribution has two parts. First, a small continuous-charging construction makes the obstruction explicit under full terminal replenishment. A perturbation permits a positive reserve, charging losses and one finite connector. A replication calculation then separates an aggregate planning gap from the incentives of a whole-fleet operator and of participants with individually

reserved resources. A convex hourly generator model adds a separate ramp-constrained example and a support test allowing nonunique dispatch prices. Dual decomposition is interpreted through the fleet hull, with explicit limits on convergence claims. These are application-specific results within established nonconvex pricing theory. Second, an energy argument strengthens the public timetable’s lower bounds, and existing computations are organized around physical planning cost, hull cost and own-price regret. This makes the computational evidence answer the same question as the analytical model.

The computational scope comprises small synthetic diagnostics, an exact generation-dispatch extension, and two depot variants of one public timetable. The public bounds remain open. A separate development supplement preserves exploratory method comparisons, including unsuccessful runs. Broader computational validation remains part of the research program.

2 Related work

Electric-bus scheduling jointly assigns service to vehicles and tracks battery and charging decisions. Wu et al. study multi-depot scheduling with grid objectives; de Vos et al. model partial charging and station capacity; Parmentier et al. scale route and recharge scheduling by column generation [Wu et al., 2022, de Vos et al., 2024, Parmentier et al., 2023]. These works establish operational roles for duty construction, energy limits, and charging resources. Löbel et al. model nonlinear charging with variable recharge rates, showing why charging approximations require care [Löbel et al., 2024]. Here we ask how a complete fleet plan relates to the convex hull of complete plans under a common convex supply-cost model. A duty-column relaxation need not equal this whole-fleet hull, and a mean load in the hull need not be executable as one day’s bus schedule.

Aggregating charging flexibility into a market bid is an established application. Zoltowska and Lin use bus-specific availability and energy inputs, mixed-integer aggregation and disaggregation, and a linear hourly auction representation [Zoltowska and Lin, 2021]. Our distinction is the additional choice of discrete service assignments and whole-fleet operating cost: support of an aggregate charging allocation does not automatically settle support over those alternatives. This is a scope distinction, not a claim that their approach ignores discrete charging constraints. Public timetable data from Sistig et al. provide operational inputs for testing such questions [Sistig et al., 2025a,b].

Nonconvex electricity-market research explains why marginal prices can fail to support dispatch with indivisible operating decisions. Gribik et al. compare unit-commitment pricing rules and discuss energy uplift when no energy-price vector supports selected quantities [Gribik et al., 2007]; O’Neill et al. construct prices from a mixed-integer model and associated linear program [O’Neill et al., 2005]. Convex-hull pricing studies uniform prices through convexification: Schiro et al. discuss its formulation and implementation, and Hua and Baldick give a convex primal formulation [Schiro et al., 2016, Hua and Baldick, 2017]. Madani et al. compare convex-hull, integer-programming, and European-style pricing with nonconvex demand bids; Andrianesis et al. compute convex-hull prices by Dantzig–Wolfe decomposition and column generation [Madani et al., 2018, Andrianesis et al., 2022]. Hümbes et al. characterize linear-price support for decentralized problems with integralities and complete linear feasible-set descriptions [Hümbes et al., 2022]. Bichler et al. study pricing with flexible demand in nonconvex electricity markets and the tradeoffs between pricing objectives [Bichler et al., 2023]. These are precedents for convex-hull pricing and price-support questions; we use the same general supporting-price logic for a stated fleet set and supply-cost function. Our own-price best-response test is conditional on that model, not a general

market-clearing theorem or settlement design; fleet regret is not an uplift payment under these market rules.

Aggregation results require a scale and a participant definition. Starr’s quasi-equilibrium result gives the classical perspective for nonconvex economies [Starr, 1969]. Hreinsson et al. apply Shapley–Folkman reasoning to the action sets and costs of aggregated dispatchable demand and develop reduced convex representations for system operators [Hreinsson et al., 2021]. Our replication calculation is explicit: its planning gap tends to zero while a whole operator’s absolute regret can stay positive along a subsequence, although per-bus and reserved-pair incentives vanish. Approximate convexity of an aggregate does not, by itself, bound the absolute deviation gain of a composite participant.

Alizadeh et al. couple transportation and power networks using continuous traffic path flows, generator dispatch, locational prices and dual decomposition [Alizadeh et al., 2017]. Scaglione derives marginal prices for generation trajectories with intertemporal ramp constraints [Scaglione, 2016]. These motivate our generation-side dispatch extension and price-coordination interpretation. Their convex flow or dispatch formulations do not establish support for an atomic fleet with discrete service assignments. Gan et al. likewise coordinate continuous EV charging profiles with fixed availability and energy requirements [Gan et al., 2013]; bus assignments additionally change charging windows and the joint feasible load set. We examine what survives this change in the participant’s feasible set, rather than attributing generic nonconvergence to those convex coordination methods.

Fixed-schedule fleet counting has a classical analogue in Dantzig and Fulkerson’s minimum-tanker problem [Dantzig and Fulkerson, 1954]. Such flow reductions count vehicles covering fixed movements, but do not certify battery states, shared-charger feasibility, or price support. We separate minimum-fleet flow diagnostics, charging replay, and convex-hull economic comparisons.

3 Complete fleets and marginal-price support

The unit of optimization is a *complete* fleet plan, rather than an individual vehicle duty. Fix a timetable, allowed directed movements, a charging horizon, and the equipment and energy assumptions. Let $\mathcal{X} \neq \emptyset$ contain precisely the plans that cover every mandatory service once, assign the services to used buses, route those buses through allowed movements, respect battery and charger constraints, and restore every used bus to its required terminal inventory. For $x \in \mathcal{X}$, write $c(x)$ for intrinsic operating cost and $e(x) \in \mathbb{R}_+^T$ for grid energy bought in the T market intervals. An unused bus contributes neither charging nor cost. We use the same $\mathcal{X}$, c , and e for physical planning, price response, and convexification. Finitely many discrete duties with bounded continuous charging give a compact $\mathcal{X}$ in the cases below; assume more generally that its image $\{(e(x), c(x)) : x \in \mathcal{X}\}$ is compact.

In the timed physical model, each used path starts at battery inventory B . A service or directed movement consumes its declared energy in chronological order. Inventory remains between reserve r and capacity B at every event. A depot charging session buying q grid kWh adds ηq inventory, where $0 < \eta \leq 1$, within its availability window and bus-power limit. Concurrent sessions obey the shared power or connector limit. Each path finishes at inventory B by the declared horizon. In the homogeneous single-connector formulation, charging intervals are split at all visit and market boundaries. If interval k has duration d_k and power P_k , its charges obey $0 \leq z_{mk} \leq P_k d_k y_m$ and $\sum_m z_{mk} \leq P_k d_k$, where y_m indicates the selected visit. Serial allocation of z_{mk}/P_k hours to the visits realizes these totals when switching time is zero, efficiency is constant, and charging does not

taper. The worked constructions below assume zero travel cost and no unlisted travel or auxiliary load; the public case declares its movement energy separately.

Let $F : \mathbb{R}^T \rightarrow \mathbb{R} \cup \{+\infty\}$ be a proper, closed, convex supply cost, finite at every physical load. Its effective domain includes the feasible loads but may also encode generation capacity and ramp limits. Write $\mathcal{E} = \text{conv}\{e(x) : x \in \mathcal{X}\}$. For the support equivalence, assume either $\text{ri } \mathcal{E} \cap \text{ri}(\text{dom } F) \neq \emptyset$, or that F has a finite convex minorant on $\mathbb{R}^T$ that agrees with it on $\mathcal{E}$. The latter covers the nonnegative-domain quadratic used below via its unrestricted quadratic formula. Here F represents the supplier's aggregate incremental provision cost for fleet charging, conditional on background demand already accounted for. A system operator posts the marginal tariff calculated from a candidate or forecast fleet load before the fleet chooses a schedule. The tariff is held fixed when evaluating alternative complete fleets. This ex-ante pricing rule is the institution tested below; the supplier is not a strategic player and the fleet does not anticipate a tariff reset during its deviation. The physical and complete-fleet-hull values are

$$\begin{aligned} D &= \min_{x \in \mathcal{X}} \{c(x) + F(e(x))\}, \\ CH &= \min_{(e,c) \in \text{conv}\{(e(x), c(x)) : x \in \mathcal{X}\}} \{c + F(e)\}, \quad \Delta = D - CH \geq 0. \end{aligned} \quad (1)$$

The hull mixes whole feasible plans, including their charger constraints; its mean load is a pricing and bounding object, not necessarily a dispatch. For a posted price p , define the complete-fleet response $V(p) = \min_{x \in \mathcal{X}} \{c(x) + p^\top e(x)\}$. A marginal price at a physical plan x is any $p \in \partial F(e(x))$; this set can contain more than one price and need not be nonempty at every boundary load. Define

$$r(x; p) = c(x) + p^\top e(x) - V(p). \quad (2)$$

For a stated price selection p_x , write $r(x) = r(x; p_x)$; a nonsmooth model must declare that selection when reporting own-price regret. The alternative in $V(p)$ takes the posted price as fixed, as the stated tariff rule requires. A strategic fleet instead compares bills such as $c(y) + p_y^\top e(y)$ across its alternatives y and anticipates the resulting price change. That objective differs from both $c(y) + p^\top e(y)$ and the planner's $c(y) + F(e(y))$; it needs a specified settlement rule and strategic game. The present price-support and compensation statements make no claim about strategic behavior [O'Neill et al., 2005, Gribik et al., 2007, Gan et al., 2013].

Proposition 1 (Complete-fleet support). *Under the common-model and attainment assumptions above, every physical $x \in \mathcal{X}$ and every $p \in \partial F(e(x))$ satisfy*

$$r(x; p) \geq c(x) + F(e(x)) - CH \geq D - CH = \Delta. \quad (3)$$

Moreover, $D = CH$ if and only if some physical plan is a best response at some price in its subdifferential. If $D = CH$, every physical planner optimizer is supported by every attained maximizer of the hull dual $V(p) - F^(p)$.*

The proof in Appendix A is a specialization of classical nonconvex price-support and convexification arguments [Starr, 1969, O'Neill et al., 2005, Gribik et al., 2007, Hua and Baldick, 2017, Schiro et al., 2016]. It says nothing about whether an iterative price adjustment converges or which optimum an algorithm selects. The stated qualification ensures strong Fenchel duality and an attained supporting price. If an economic price convention admits only a subset of $\partial F(e(x))$, the converse additionally requires an attained supporting price in that subset. In particular, a normal caused solely by extending a nonnegative-load cost as $+\infty$ to negative loads is not automatically a generation price.

The accounting identity also requires a stated deviation right. In this paper one fleet controls all buses and its alternative must be a member of the same $\mathcal{X}$. Separate operators using shared chargers would need reserved capacities, trades, or congestion rights before a unilateral opportunity set could be defined. For the single fleet, put $F^*(p) = \sup_{u \in \mathbb{R}^T} \{p^\top u - F(u)\}$ and define

$$\text{LOC}_{\text{fleet}}(x; p) = c(x) + p^\top e(x) - V(p), \quad (4)$$

$$\text{LOC}_{\text{supply}}(x; p) = F(e(x)) - p^\top e(x) + F^*(p). \quad (5)$$

Whenever $F^*(p)$ is finite, both terms are nonnegative and their sum is $c(x) + F(e(x)) - [V(p) - F^*(p)]$. At a hull-optimal price and a physical planner optimizer, strong Fenchel duality makes that sum Δ ; the fleet and supply portions depend on the chosen common price. At any $p \in \partial F(e(x))$, supply LOC is zero and fleet LOC equals $r(x; p)$. No budget-balance or payment institution follows from this arithmetic.

For quadratic supply $F(e) = a^\top e + \frac{1}{2}e^\top Qe$ on $\mathbb{R}_+^T$ with $Q \succeq 0$, take the unrestricted-quadratic gradient price $p_x = a + Qe(x)$ and write $\|z\|_Q = (z^\top Qz)^{1/2}$. There is a useful complement to the lower bound. Let $R_Q = \max_{x, y \in \mathcal{X}} \|e(x) - e(y)\|_Q$ and $H(x) = c(x) + F(e(x)) - CH$. Then

$$H(x) \leq r(x) \leq H(x) + R_Q \sqrt{2H(x)}. \quad (6)$$

Thus a small cost gap controls own-price regret only together with the curvature-weighted range of feasible fleet loads. The proof permits singular Q and boundary loads and appears in Appendix A.

For numerical certification, valid bounds $D \in [L_D, U_D]$ and $CH \in [L_{CH}, U_{CH}]$ imply $\Delta \in [L_D - U_{CH}, U_D - L_{CH}]$. A replay-feasible physical plan supplies an upper bound on D ; a mixture of replay-feasible complete plans supplies an upper bound on CH . At any price with finite $F^*(p)$, a global price-response lower bound $\underline{V}(p) \leq V(p)$ gives the hull lower bound $\underline{V}(p) - F^*(p)$. A nearly solved restricted column pool alone cannot supply that global lower bound. In the separable implementation, $F(e) = \sum_t (a_t e_t + b_t e_t^2 / 2)$ for $e \geq 0$ and $b_t \geq 0$; the conjugate term is $[p_t - a_t]_+^2 / (2b_t)$ when $b_t > 0$, and is finite at $b_t = 0$ only for $p_t \leq a_t$. These are standard convexification and Fenchel certificates over complete plans, rather than a new general convex-hull pricing method [Hua and Baldick, 2017]. Section 7 compares two specified market states.

4 Exact replenished-fleet constructions

Consider two mandatory depot-to-depot services. Service A occupies hour $[0, 1]$ and service B occupies $[2, 3]$; each consumes 15 kWh. At most two identical buses may be used. Each used bus starts at its full 20 kWh capacity and must return to 20 kWh by hour 4. The early charging window is $[1, 2]$, with 10 kW per bus and in total. The terminal window is $[3, 4]$, with 30 kW per bus and in total and enough connectors. Charging is continuous and lossless; reserve, travel cost, deadhead energy, auxiliary load, switching time, and taper are zero. No charging is available outside these windows. The terminal charge is physical replenishment, so its energy and supply cost are included.

There are only two unlabeled service partitions. If one bus covers A and B, it reaches 5 kWh after A and needs at least 10 kWh before B. The early power limit fixes its early grid energy at $x = 10$ kWh; it buys 20 kWh terminally. If two buses cover one service each, the A bus buys any $x \in [0, 10]$ kWh early and $15 - x$ kWh terminally, while the B bus buys 15 kWh terminally. Every used bus returns full. Both structures therefore buy exactly 30 kWh, and initial inventories cannot subsidize the horizon.

Let a used bus cost $f = 7$ synthetic units and let grid supply cost be

$$F(x, z) = 4x + \frac{x^2 + z^2}{10}, \quad x, z \geq 0. \quad (7)$$

The one-bus physical cost is $7 + F(10, 20) = 97$. The two-bus branch has cost $14 + F(x, 30 - x)$, minimized at $x = 5$ with value 99. Yet the complete-fleet hull has a lower objective.

Proposition 2 (Replenishment does not ensure price support). *For this physical model,*

$$D = 97, \quad CH = \frac{7591}{80} = 94.8875, \quad \Delta = \frac{169}{80} = 2.1125. \quad (8)$$

The physical planner optimum has own-price regret 13. At a common hull-optimal price its fleet lost opportunity is zero and its supplier lost opportunity is $169/80$.

The hull optimizer mixes the one-bus plan $(x, z) = (10, 20)$ with weight $27/40$ and the two-bus plan $(0, 30)$ with weight $13/40$. Its mean early load is $27/4$ kWh. Both components are physical; their mean load and mean operating cost are a convexification, not a fractional bus schedule. Running the two plans on different days gives mean system cost 99.275, because averaging physical supply costs differs from evaluating the convex supply cost at their mean load. The exact hull and price calculations are in Appendix A.

At the physical optimum, the own marginal price is $(6, 4)$ synthetic units/kWh. The one-bus private objective is 147, while the two-bus plan charging entirely terminally has private objective 134. The resulting 13-unit regret is a fixed-price incentive to change the whole fleet. The hull price is $(5.35, 4.65)$: the two plans supporting its mean tie at private objective 153.5. At that price, the physical one-bus optimum has no fleet lost opportunity, but the supplier term is $169/80$. These two diagnostics use different prices and answer different questions.

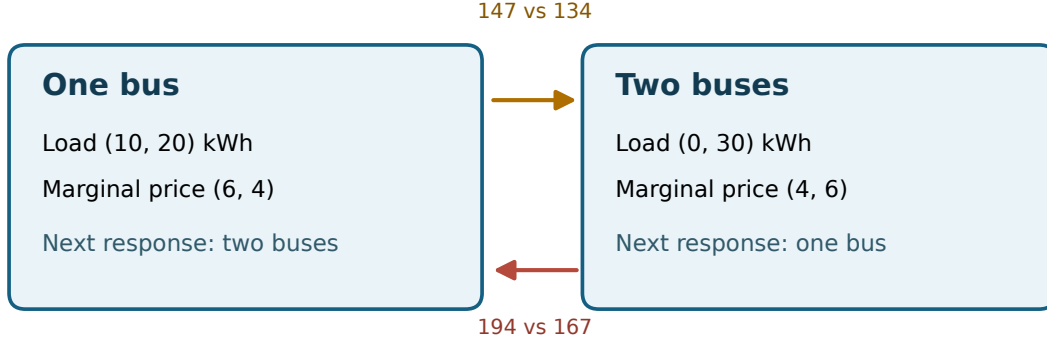
Starting the undamped best-response iteration at the one-bus load gives the exact two-cycle in Figure 1. At $p = (6, 4)$, one bus pays 147 and the two-bus fleet with load $(0, 30)$ pays 134, so the response is two buses. Updating the same supply gradient at that load gives $p = (4, 6)$; now the current two-bus plan pays 194, while the one-bus fleet with load $(10, 20)$ pays 167. Even the cheapest two-bus alternative at this price costs 174. Both schedules fully replenish their buses. The iteration alternates without damping or tie breaking. It illustrates this posted-price rule, not a general claim about price-update algorithms.

The binding 10 kW early limit is a fragile nominal parameter: at the same 20 kWh capacity, a positive reserve or charging loss can remove the one-bus branch. A nearby construction allows those constraints while retaining a gap. Keep two 15 kWh services, battery capacity 20, and (7), but use reserve $r = 1$ kWh, grid-to-battery efficiency $\eta = 19/20$, 12 kW individual and shared early power, and a single 30 kW terminal connector. The one-bus plan buys $220/19$ grid kWh early and 20 terminally, with battery states $20 \rightarrow 5 \rightarrow 16 \rightarrow 1 \rightarrow 20$. Its early power has $8/19$ kWh of headroom. The supporting two-bus plan buys $30/19$ early and 30 terminally; the two terminal sessions take $9/19$ and $10/19$ hour and fit sequentially on the single connector. Every used bus again finishes full.

Proposition 3 (A positive-gap neighborhood). *Under the modified parameters,*

$$D = \frac{38527}{361}, \quad CH = \frac{2987911}{28880}, \quad \Delta = \frac{94249}{28880} > 0. \quad (9)$$

With the 30 kW terminal connector fixed, strict feasible-headroom and optimality inequalities preserve a positive gap in an open neighborhood of the stated reserve, efficiency, and early power.



At each step: best response to the posted price, then update price at the new load

Figure 1: Exact undamped best-response cycle in the nominal two-service model. Initialize at the one-bus physical optimum; each arrow holds the displayed price fixed for the response, then updates to the marginal price of the new load. Numbers on arrows compare the current fleet’s bill with the responding fleet’s bill, in synthetic units.

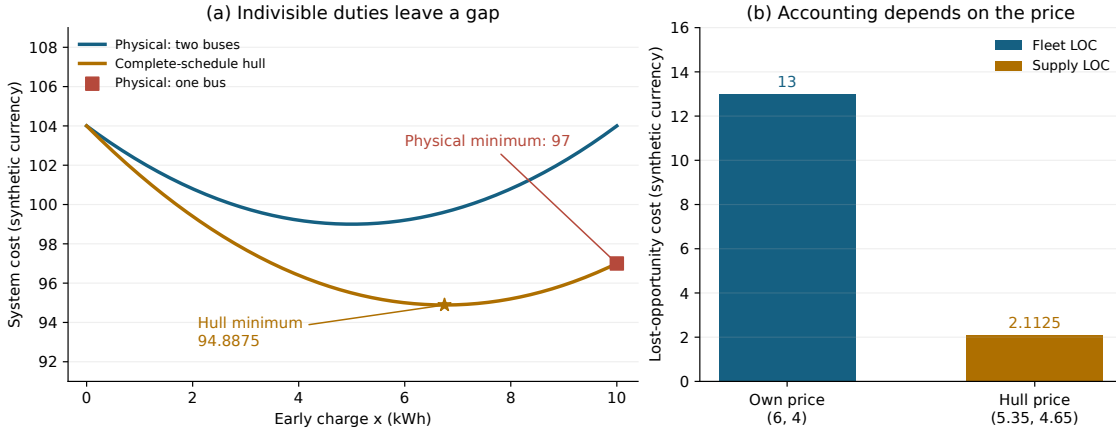


Figure 2: Exact physical branches, complete-fleet hull, and lost-opportunity accounts in the synthetic two-service case. Prices are synthetic units/kWh; the hull mean is not an executable daily fleet.

The interval and strict-inequality proof is in Appendix A. The neighborhood is conditional on this hardware model’s zero switching time, constant efficiency, and lack of taper. For perspective on connector adequacy, if the original lossless case has one terminal connector of power $K \leq 30$ kW, it is infeasible for $K < 20$, has zero gap for $20 \leq K \leq 23.5$, and has a positive gap for $23.5 < K \leq 30$. Thus the presence of one connector alone says nothing about price support.

4.1 Replication and the participant scale

Replicate A and B to n simultaneous copies each. Give the fleet n early 10 kW connectors and n terminal 30 kW connectors. Battery and service energy per bus stay fixed; supply cost scales with demand as

$$F_n(E, L) = 4E + \frac{E^2 + L^2}{10n}. \quad (10)$$

If m buses each cover an A–B pair, $n - m$ A-only and $n - m$ B-only buses complete the services. There are $2n - m$ used buses, and physical early energy ranges over $[10m, 10n]$. Terminal sessions have an explicit connector-feasible assignment: paired buses take m connectors, while each A-only/B-only pair can share one of the other $n - m$ terminal connectors sequentially. All buses return full.

Proposition 4 (A vanishing gap with aggregate regret). *The complete projected hull has lower operating-cost boundary $14n - 0.7E$ and optimum $CH_n = 7591n/80$. Every physical optimum takes an integer m nearest to $27n/40$; both neighboring integers are retained at a tie. With $\delta = m - 27n/40$,*

$$\Delta_n = \frac{20\delta^2}{n} \leq \frac{5}{n}. \quad (11)$$

Along $n = 40k + 1$, the whole-operator own-price regret is

$$r_n = \frac{351}{40} + \frac{169}{40n} \longrightarrow \frac{351}{40}, \quad \Delta_n = \frac{169}{80n} \longrightarrow 0. \quad (12)$$

For multiples of 40, both quantities are exactly zero.

The operator in (12) can rearrange the whole growing fleet while taking its posted price as fixed. Its regret per used bus, and regret divided by physical cost, vanish. A different institution gives each of n operators one A/B pair plus a reserved early and terminal connector. Each then chooses either the one-bus plan or a two-bus plan with $x \in [0, 10]$; these separate deviations are physically compatible under the reserved rights. At the original fleet optimum’s marginal price, each pair’s regret is at most $20/n$, and their sum equals the whole-fleet regret in this symmetric construction. Along $n = 40k + 1$ a dissatisfied one-bus operator has regret $13/n$, although the aggregate tends to $351/40$. The scaling example consequently distinguishes absolute cost gap, whole-operator incentive, and per-participant incentive without asserting a result for arbitrary shared-charger games. At $n = 20$, the two tied planner optima have distinct own-price regrets, 7 and 14.

Shapley–Folkman aggregation arguments, as used for nonconvex power-market demand by Hreinsson et al. [2021], concern sums of separately bounded participant sets in a fixed coupling dimension. They can bound aggregate convexification effects as the number of sufficiently small participants grows. The $5/n$ bound here instead follows from the exact geometry and cost scaling of this replicated family. Its persistent $O(1)$ absolute whole-operator regret is likewise a property of this family’s single operator and posted-price deviation set, not a general consequence of the Shapley–Folkman theorem. Per-pair regret refers only to the separate reserved-resource institution just defined.

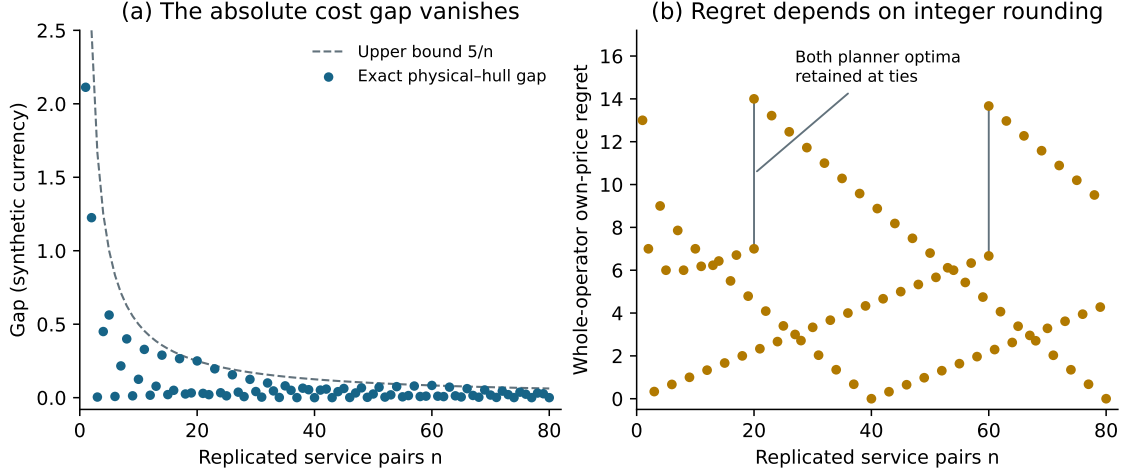


Figure 3: Exact replication for $n = 1, \dots, 80$, including both physical optima at ties. The planning gap follows the $5/n$ bound while whole-operator own-price regret varies with integer rounding. Values are synthetic units, not observations from a bus system.

5 Generation dispatch and price coordination

5.1 An hourly convex supply model

Let b_t be background demand and L_t fleet energy in one-hour intervals. With linear marginal costs a_{it} , generator limits $\bar{g}_{it}$, and declared initial outputs g_{i0} , define

$$\begin{aligned}
 H(d) = \min_g \quad & \sum_{i,t} a_{it} g_{it}, \\
 \text{subject to} \quad & \sum_i g_{it} = d_t, \quad 0 \leq g_{it} \leq \bar{g}_{it}, \\
 & -R_{it}^- \leq g_{it} - g_{i,t-1} \leq R_{it}^+.
 \end{aligned} \tag{13}$$

Piecewise-linear or convex quadratic production costs give analogous convex models. The incremental supply cost is $G(L) = H(b + L) - H(b)$, identified with F in Section 3. The baseline is fixed across alternatives. Balance multipliers are marginal prices; ramps couple their determination across time. This hourly model follows the dispatch-price interpretation of Scaglione [2016], without claiming to reproduce that paper’s continuous-time trajectory model. No generator commitment binaries or network congestion are included. Nodal balances and transmission constraints would require depot-specific loads and locational prices.

All physical fleet loads in the example below are supply-feasible. More generally, capacity limits require an explicit supply-feasible deviation set or a specified backup-supply rule. Extending a cost as $+\infty$ cannot silently change which fleet alternatives are admissible. A generation LP may have several optimal balance multipliers. Report a selected price and, when testing existence of support, consider the full declared admissible price set. One solver-selected price alone cannot exclude the others.

5.2 What dual coordination does and does not imply

Dualize $L = e(x)$ in joint fleet and supply planning. With $V(p) = \min_{x \in \mathcal{X}} \{c(x) + p^\top e(x)\}$, the dual is

$$\max_p Q(p), \quad Q(p) = V(p) - G^*(p). \quad (14)$$

Under Proposition 1’s qualification its value is CH , so the physical primal–dual gap is exactly $D - CH$. This is the classical convex-hull pricing connection [Gribik et al., 2007]. Alizadeh et al. [2017] coordinate convex traffic flows and power dispatch through dual prices. Replacing those divisible flows by an atomic complete-fleet oracle retains a concave dual but need not retain zero physical duality gap. Complete-schedule column generation provides another route to the same convexification [Andrianesis et al., 2022].

At p_k , let x_k globally minimize the fleet objective and let s_k maximize $p_k^\top s - G(s)$. Then $e(x_k) - s_k$ is a supergradient of Q , and a projected ascent step is

$$p_{k+1} = \Pi_P\{p_k + \alpha_k[e(x_k) - s_k]\}. \quad (15)$$

Suppose a closed convex price region P contains an unrestricted dual optimizer p^* , the oracles attain their optima, and the supergradient norms are at most M along the run. The projection inequality gives

$$Q(p^*) - \max_{1 \leq k \leq K} Q(p_k) \leq \frac{\|p_1 - p^*\|^2 + M^2 \sum_{k=1}^K \alpha_k^2}{2 \sum_{k=1}^K \alpha_k}. \quad (16)$$

Thus $\alpha_k \propto k^{-1/2}$ gives a best-dual-value guarantee of order $\log K / \sqrt{K}$. It does not alone prove convergence of each price iterate, a unique limiting price, or a physical fleet. A price restriction excluding all unrestricted optimizers instead solves a different dual. An inexact fleet incumbent is not an exact supergradient oracle.

A balanced fleet and supply response at one common price would attain both primal and dual values and force $D = CH$. A positive gap therefore excludes a balanced joint limit of exact responses at a finite limiting price. It does not imply that every schedule sequence must switch forever: a tie rule can repeatedly choose one fleet at a hull price while its load still differs from the supply response. The undamped gradient-reset cycle in Figure 1 is a different update from (15).

For the quadratic construction, we ran (15) for 20,000 analytic oracle calls, starting at $(6, 4)$, projecting onto $\mathbb{R}_+^2$, and taking $\alpha_k = 0.5/\sqrt{k}$. All continuous two-bus charging choices are minimized analytically; fleet ties select one bus. Before call 20,000, the price is approximately $(5.3412, 4.6588)$, the current dual is 94.8300, and the best dual so far is 94.887499, compared with exact $CH = 94.8875$. The one-bus frequency is 0.6682 versus the hull weight 0.675; the step-weighted frequency is 0.6603. The last 60 responses contain 38 bus-count switches. Figure 4 distinguishes current from best dual bounds. These are cheap analytic calls and finite-run diagnostics, not 20,000 timetable MIPs or a proof of primal convergence.

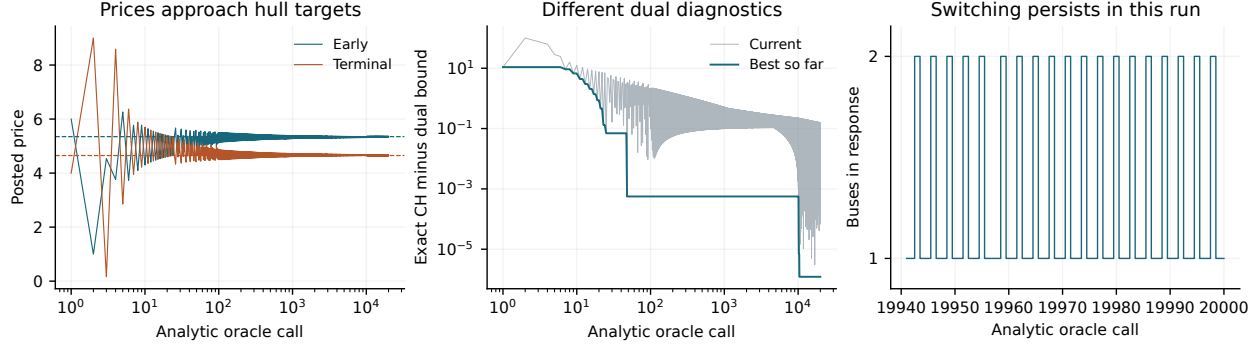


Figure 4: Projected dual ascent on the existing quadratic two-service example, with the declared initialization and step sizes. Dashed lines are the exact hull-price coordinates. A nearly optimal best dual bound coexists with changing physical responses and a less accurate current dual bound.

5.3 An exact two-generator example with hourly ramps

Keep every fleet constraint and continuous charging choice from Section 4. Generation now spans the consecutive hours $[1, 2]$, $[2, 3]$, and $[3, 4]$: the fleet load is $L = (x, 0, 30 - x)$ and background demand is $b = (0, 15, 0)$. The middle hour, when service B operates, cannot be omitted from an hourly ramp model. Both generators start at zero; no terminal generation level is imposed. Costs are declared synthetic hourly offers, with no fixed commitment costs. Generator A has marginal costs $(7, 5, 1)$ and capacities $(10, 15, 15)$; generator B has costs $(6, 5, 5)$ and capacities $(10, 15, 30)$. A's initial upward limit is 10 and its two subsequent hourly upward limits are 5. All downward limits and B's upward limits are 30. Power and energy numbers coincide in these one-hour intervals.

Background supply costs $H(b) = 75$. Let a, m, z denote A's early, middle and terminal outputs. On the full feasible fleet interval $0 \leq x \leq 10$, subtracting the baseline leaves $150 + x + a - 4z$. The two ramps and terminal capacity imply $z \leq \min(15, a + 10)$, attained with $m = a + 5$ for every $a \in [0, x]$. Minimizing over a therefore gives the exact supply value

$$G(x, 0, 30 - x) = \begin{cases} 110 - 2x, & 0 \leq x \leq 5, \\ 95 + x, & 5 \leq x \leq 10. \end{cases} \quad (17)$$

The one-bus plan costs $7 + G(10, 0, 20) = 112$; the best two-bus plan costs $14 + G(5, 0, 25) = 114$. The complete-fleet hull retains the operating-cost boundary $14 - 0.7x$, giving

$$D_G = 112, \quad CH_G = \frac{221}{2} = 110.5, \quad D_G - CH_G = \frac{3}{2}. \quad (18)$$

The minimizing mixture gives equal weight to the one-bus load $(10, 0, 20)$ and the two-bus load $(0, 0, 30)$. At its mean $(5, 0, 25)$, dispatch is $A = (5, 10, 15)$ and $B = (0, 5, 10)$, with both of A's hourly upward ramps binding (Figure 5).

At that mean the exact set of balance prices is $p = (s, 5, 5)$, $3 \leq s \leq 6$. Positive B output fixes the last two prices; A's stationarity gives equal ramp rents $7 - s$ and terminal-capacity rent $s - 3$, while B's unused early output requires $s \leq 6$. Selecting $p^* = (5.7, 5, 5)$ makes the two supporting fleet plans tie at private cost 164. Its generator dual supplies the global cut $G(L) \geq (p^*)^\top L - 53.5$, tight at the mean. Thus the dual lower bound is $164 - 53.5 = 110.5$, equal to the mixture upper bound. This derivation is independent of the quadratic example's prices.

The physical one-bus optimum has the unique dispatch price $(6, 5, 5)$ and regret 3. The physically feasible two-bus plan at the kink $x = 5$ instead has a price-dependent regret: $32 - 5s$ for

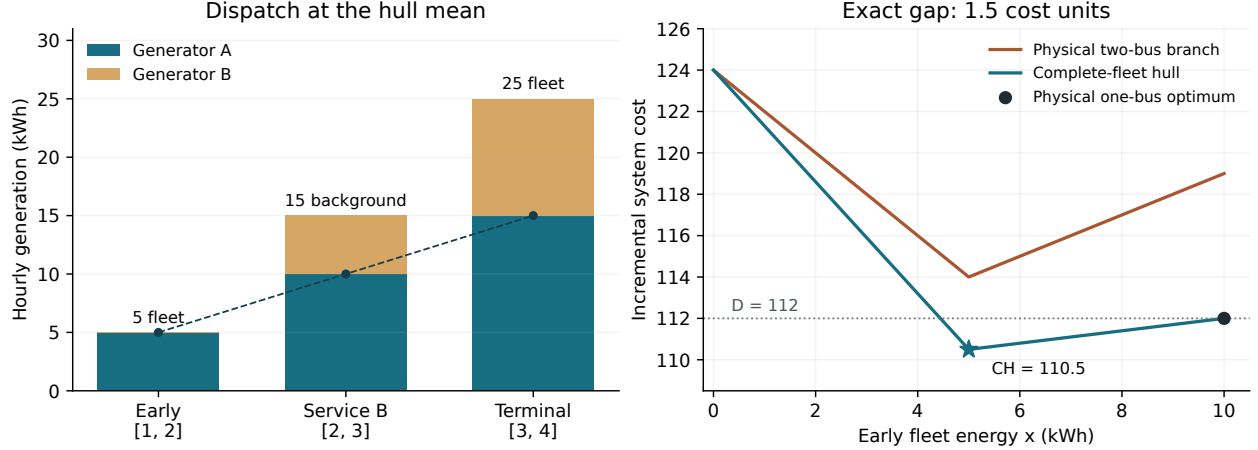


Figure 5: The hourly generation example. Left: generation serving the hull mean; the middle hour serves only background demand, and A rises by its 5-unit limit in each interval. Right: physical two-bus and hull costs over the full continuous charging interval, with the one-bus optimum marked separately. All costs and offers are synthetic.

$3 \leq s \leq 5.7$, and $5s - 25$ for $5.7 \leq s \leq 6$. Its smallest admissible regret is 3.5, attained at $s = 5.7$, equal to its physical cost minus CH_G . This illustrates why a nonsmooth supply model needs a declared price selection or a test over the admissible set.

Relaxing only A’s two inter-hour upward limits from 5 to 30 gives $G_0(x, 0, 30 - x) = 90 + x$. Both the physical and hull minima then equal 104, attained by the two-bus plan with $x = 0$. These controlled synthetic cases isolate the change in the supply constraints; they are not measurements of price-support failure in a public timetable. Small LP solutions independently check dispatch, balance prices and hull values, while the affine branch derivation and rational dual certificate establish the exact claims.

6 A public timetable: feasible fleets and bounded costs

The exact two-service example isolates the mechanism; it does not measure its prevalence or size in service. For a timetable-scale check we use the published Hildenbrand instance from the Sistig electric-bus scheduling data [Sistig et al., 2025a,b]: 37 mandatory services with a complete directed deadhead matrix. We form two *separate* one-depot cases, using depot 15 or depot 16. These are declared modifications of the source network, not a reconstruction of its two-depot, multiple-output charging design. Timetable and directed travel times retain their source offsets, including the next-day services. Matrix travel times depend on origin and destination, not on departure time.

Energy and monetary inputs are explicit scenario assumptions, not observed vehicle telemetry or source objective coefficients. Service energy is 1.20 kWh/km plus 16 kWh per service hour. Directed deadheads use 0.96 kWh/km plus 16 kWh per travel hour, and explicit off-depot waiting uses 16 kWh/h; depot-parked auxiliary draw is zero. Each used bus has 400 kWh usable inventory, zero additional reserve, unit charging efficiency, and a 360 kW acceptance limit. One 360 kW connector is shared at the selected depot. All used buses start full and return full by a synthetic 30:00 service-block boundary. Charging has no taper and no switching time. This finite block does not establish a repeating daily operation. A 37-bus one-service-per-bus construction confirms feasibility of both cases, but has no optimization role. Monetary travel cost is zero: the objective

charges the used-bus fee and grid energy, while movement energy and time remain in the physical constraints.

We first use synthetic used-bus cost $f = 100$ and flat grid price $a = 0.2$ per kWh. Because this objective is linear in load, minimizing it over the hull of complete plans has the same value as minimizing it over physical plans; this flat-price experiment cannot exhibit a physical-hull gap. The original fixed-budget native runs returned physically replayed two-bus schedules in both depot cases without proving objective optimality. A separate, already archived round-0 solve within the later nonlinear planner used the same complete depot-15 physical case and a tangent objective that reduces exactly to the flat price. That call returned a *native numerical* optimum of approximately 408.53, with matching incumbent and lower bound under solver tolerances. This is a retrospective interpretation of one call, not a new flat-price experiment or a reclassification of the nonlinear campaign. The numerical certificate does not identify the exact rational-model optimum. The following ideal stored-input certificates answer that distinct exact-arithmetic question.

For both depot graphs, a one-bus path is impossible. Chronologically forced adjacent services leave no admissible depot visit, while mandatory service energy alone exceeds one bus’s 400 kWh inventory. An exact cardinality-gated path-cover relaxation of each full 37-service graph gives flat-objective lower bounds displayed outward as 404.92 at depot 15 and 414.39 at depot 16. The bounds retain every declared connection mode but relax some physical constraints. Separately, an exact rational depot-15 witness covers all 37 services with two buses, respects the shared connector, and restores both batteries to 400 kWh by 30:00. Hence depot 15’s ideal model has a minimum fleet of two and the *flat-cost optimum* lies in

$$D_{15}^{\text{flat}} \in [404.92, 408.54] \quad (\text{outward-rounded exact rational bounds}). \quad (19)$$

The bounds do not identify the optimum. Depot 16 has the exact at-least-two obstruction and lower bound, but its returned two-bus upper witness remains numerical, yielding a mixed-evidence interval $[414.39, 433.75]$ after outward rounding. In particular, a native solver’s rounded matrix and its feasibility tolerances are distinct from the ideal stored-input rational model.

For a nonlinear illustration on *the same finite depot-15 ideal model*, let e_t be hourly grid energy for 30 market intervals and set

$$F(e) = \sum_{t=1}^{30} \left(0.2e_t + \frac{e_t^2}{1800} \right). \quad (20)$$

The coefficients and bus cost remain synthetic. The exact two-bus witness gives a physical upper bound displayed as 512.77. To bound the complete hull from below, let L_{flow} denote the exact depot-15 two-path flat-cost relaxation value. Because monetary travel cost is zero, it implies at least $E_{\text{min}} = (L_{\text{flow}} - 2f)/a \approx 1024.62$ kWh for any two-bus plan. One bus is impossible, and a uniform-price comparison using the service-energy floor excludes three-or-more-bus plans from improving the dual lower certificate. Restricting the Fenchel inequality to the 24 hours when charging can occur gives the exact rational lower bound displayed as 429.22; Section 7 derives the charging-window cap. Therefore

$$429.22 \leq CH_{15} \leq D_{15} \leq 512.77, \quad 0 \leq D_{15} - CH_{15} \leq 83.55. \quad (21)$$

All displayed endpoints are rounded outward from the exact bounds. They do not establish CH_{15} or D_{15} individually, nor a strictly positive gap. The public case thus demonstrates physically feasible complete-fleet witnesses and useful bounds under declared assumptions, while leaving public price support open.

7 Computational price-support diagnostics

The computational question is whether a physical optimum is supported by its own marginal price. A useful diagnostic therefore reports the physical value D , hull value CH , their nonnegative difference and own-price regret together. We combine analytical public lower bounds with archived development-screen results. The two depot variants represent one public timetable.

7.1 An analytical baseline for both markets

Use the public physical cases in Section 6. Market 0 has $a_{0,t} = 0.20$ throughout; market 1 has $a_{1,t} = 0.18$ for the first 15 hours and 0.22 for the last 15. Both have $b = 1/900$ in $F_s(e) = \sum_t (a_{s,t}e_t + be_t^2/2)$. This changes only supply cost, leaving fleet feasibility fixed. As in the existing exact public certificates, arithmetic uses the stored input coefficients as rationals; the decimal parameters here describe those inputs to their displayed precision.

Let $E_{\min}$ be the two-bus total-energy floor obtained from the exact flow relaxation, and let E_{svc} be mandatory service energy. With full replenishment, efficiency one and zero monetary travel cost, $E_{\min} = (L_{\text{flow}} - 2f)/a_0$. This gives about 1024.62 and 1071.97 kWh for depots 15 and 16, respectively; E_{svc} is about 889.12 kWh in both. The one-bus obstruction is unchanged. For a uniform price $p > 0$, every two-bus plan has $c + p \sum_t e_t \geq 2f + pE_{\min}$, whereas every plan using at least three buses has $c + p \sum_t e_t \geq 3f + pE_{\text{svc}}$. Hence, whenever

$$3f + pE_{\text{svc}} \geq 2f + pE_{\min}, \quad (22)$$

the lower bound $2f + pE_{\min}$ holds for every physical plan and its convex hull. For $p \geq \max_t a_{s,t}$, Fenchel's inequality then gives

$$CH_s \geq 2f + pE_{\min} - \sum_{t=1}^{30} \frac{(p - a_{s,t})^2}{2b}. \quad (23)$$

The full-horizon bound uses all 30 hours. The pinned physical case permits charging only on depot or terminal-return movements. Taking the union of *all* such declared windows, including movements a fleet may never select, gives no charging before minute 405.5 at depot 15 or 408.5 at depot 16. Thus the first six hourly loads are zero for every fleet. The single connector allows at most 87 and 69 kWh, respectively, in hour seven and at most 360 kWh in each later hour. Let C_t denote those outer hourly caps. Using the conjugate of supply restricted to $0 \leq e_t \leq C_t$ sharpens (23) to

$$CH_s \geq \min\{2f + pE_{\min}, 3f + pE_{\text{svc}}\} - \sum_{t=1}^{30} \sup_{0 \leq u \leq C_t} \{(p - a_{s,t})u - bu^2/2\}, \quad p \geq 0. \quad (24)$$

At the maximizing uniform test price for these four cells, the positive hourly caps do not bind. The price is $p_s^* = \bar{a}_{s,\text{active}} + bE_{\min}/24$, and the two-bus branch of the minimum applies; the three-or-more-bus expression exceeds it by 53.44–66.48 units. Exact ideal-model floors are 429.22 and 430.29 for depot 15 in original and changed markets, and 440.99 and 442.30 for depot 16. Table 1 incorporates them directly. The separate later solver-baseline comparison reported changed-market native lower endpoints 405.83 and 420.45. Its runs and timing are not pooled with the initial screen. Substituting a native flat optimum into the energy argument would make the floor native-tolerance-qualified rather than an exact ideal-model certificate.

Table 1: Physical and hull cost, gap, and incumbent fixed-price regret in both archived markets. Synthetic rows use conditional native enclosures; the nominal two-service case also has the exact values in Proposition 2. Public lower endpoints use exact ideal-model charging-availability energy floors; public upper endpoints and regret are conditional on native feasibility and bound tolerances, except for the depot-15 original-market rational witness cap. All intervals are rounded outward. Regret diagnoses the returned planner incumbent, not an optimal public fleet.

Case	Market	Physical D	Hull CH	Gap $D - CH$	Regret $r(\hat{x})$
Two-service cyclic	Original	[96.99, 97.01]	[94.88, 94.89]	[2.11, 2.12]	[12.99, 13.01]
Two-service cyclic	Changed	[98.99, 99.01]	[97.98, 97.99]	[1.01, 1.02]	[8.99, 9.01]
Three-service visits	Original	[84.45, 84.47]	[84.37, 84.47]	[0.00, 0.10]	[0.09, 0.10]
Three-service visits	Changed	[84.71, 84.73]	[84.62, 84.73]	[0.00, 0.10]	[0.09, 0.10]
Public depot 15	Original	[429.22, 512.77]	[429.22, 512.77]	[0.00, 83.55]	[266.10, 341.73]
Public depot 15	Changed	[430.29, 517.33]	[430.29, 517.33]	[0.00, 87.03]	[190.43, 235.09]
Public depot 16	Original	[440.99, 530.66]	[440.99, 530.66]	[0.00, 89.67]	[204.41, 232.22]
Public depot 16	Changed	[442.30, 547.88]	[442.30, 547.88]	[0.00, 105.58]	[215.95, 269.16]

7.2 Physical cost, hull cost and incumbent regret

The initial screen used the same two public cases and two small synthetic cases: the two-service cyclic construction and a three-service case with multiple depot visits. Each was evaluated in an original and a changed market. Table 1 shows all eight archived cells, using the screen’s physical planner, cold hull calculation and response at the returned planner incumbent’s own price, combined after the run. In the four-hour synthetic changed markets, the cyclic case uses $a_1 = (0, 3.8, 0, 0.2)$ and $b = (0, 0.2, 0, 0.2)$; the multivisit case uses $a_1 = (0.18, 0.18, 0.22, 0.22)$ and $b = (0.2, 0.2, 0.2, 0.2)$. The cyclic market has zero intercept and curvature in its service hours; only the two charging windows affect price. The archive fixes the physical inputs and detailed run budgets.

For the specific planner incumbent $\hat{x}$, let $B_{\hat{x}} = c(\hat{x}) + p_{\hat{x}}^{\top} e(\hat{x})$. A response enclosure $V(p_{\hat{x}}) \in [L_V, U_V]$ gives $r(\hat{x}) \in [B_{\hat{x}} - U_V, B_{\hat{x}} - L_V]$. The stored response price matches the gradient computed for that incumbent. We also use the response lower bound to improve the hull lower bound through $L_V - F_s^*(p_{\hat{x}})$, and any physical upper witness as a hull upper witness. The physical lower bound cannot be weaker than the hull lower bound. The table combines these same-screen inequalities before computing the gap interval. Public rows also use the exact analytical floor; depot 15’s original-market upper uses the separately archived rational witness. These are native numerical results: they retain solver and physical-replay tolerances and use rounded prices. Exact processing of their stored values does not make them ideal-model proofs.

The public regret column checks the returned schedules at their posted prices. It is not an optimal-fleet incentive estimate: planner bounds remain open and incumbent excess cost can contribute to regret. The public gap intervals all include zero.

The multivisit response supplies a hull lower bound of 84.62, reducing the same-screen planning-gap upper bound to 0.10 without another optimization. This is a consequence of combining already available certificates, not a measured benefit of reuse.

Postprocessing with the exact energy floors reduces the changed-market gap caps to 87.03 at depot 15 and 105.58 at depot 16. These are mixed-evidence bounds, conditional on the native upper witnesses representing feasible plans of the ideal model; the lower floors themselves are exact. Depot 15’s original-market row uses the stronger exact certificate from (21). None of the postprocessed intervals is a timed result of the original algorithm.

7.3 What the computations establish

Table 2: Evidence supporting the current conclusions.

Finding	Scope and remaining question
Exact support obstruction	Fully replenished examples have positive gaps under quadratic and hourly ramp-constrained supply. Replication separates the planning gap from whole-operator regret.
Dual coordination	An analytic replay improves the hull bound with switching responses. It establishes neither timetable-scale performance nor universal primal non-convergence.
Analytical public bounds	Flow, cardinality and energy inequalities give exact ideal-model lower bounds in two markets. Both depot variants share one timetable.
Incumbent diagnostic	Fixed-price responses test each named screen incumbent. The planner gap leaves optimal-fleet regret and public price support unresolved.
Repeated-solve acceleration	Development comparisons are preserved in the supplement. Cap dependence, software failures and unconfirmed start acceptance prevent a general speedup claim. Retrieval and learned proposals remain untested.

The computation presently gives public cost bounds and an incumbent response diagnostic, with the limits in Table 2. Further work must improve physical-pricing bounds, compare methods at common quality and scale to independent timetables before making general computational claims.

8 Implications and limits

The analytical and computational parts answer complementary questions. The construction proves that continuous charging and full replenishment do not remove the possibility of a marginal-price support failure. The experiments measure how difficult it is to obtain informative full-fleet bounds on declared instances. An unresolved public interval neither confirms the analytical mechanism on that timetable nor refutes it. Likewise, a strong restricted-master solution says little about the cost of discovering an ungenerated physical fleet unless global pricing has supplied a corresponding bound.

The analytical energy floor is a baseline for bounding this model. Strengthening physical pricing remains useful for narrowing global intervals. The underlying two-bus energy floor and fleet-cardinality inequality also hold for the physical price-response problem at suitable positive prices; their use in pricing must preserve the assumptions that produced them. The nonlinear hull objective bound itself does not directly constrain every linear price-response objective.

The exploratory comparisons in the development supplement are sensitive to computational limits. On a three-service case, cold runs stopped at call or arithmetic limits well before their wall-time allowance; public solver-start comparisons used one seed and did not establish whether submitted starts were accepted. Future experiments should size limits, log start acceptance, compare time to common solution quality across repeated seeds, and include independent timetables. Retained complete fleets and nearest-neighbor retrieval should precede learned proposals, with direct proposal quality, physical repair and global verification reported separately. All variants of a base timetable must stay in one data group, with independent test timetables reserved before training.

The exact two-cycle demonstrates failure of its specified undamped update. The separate dual-ascent replay improves a hull bound while physical responses keep switching during the recorded

run. Neither observation proves universal nonconvergence of atomic fleets. A positive duality gap excludes balanced optimal supply and physical fleet responses at a common price; it does not prescribe the behavior of every iteration. The generator extension uses synthetic hourly offers, and its ramp control does not establish how often this mechanism occurs in operating systems.

The physical model abstracts from stochastic delays, battery degradation, temperature-dependent consumption, nonlinear charging taper and operational switching times. Public service and travel inputs are combined with declared energy and economic assumptions; the results are not estimates of savings for the source operator. Resource rights also delimit the economic interpretation. A whole-fleet price-taking deviation, an individually reserved-pair deviation and a strategic operator anticipating its own price impact are different optimization problems. Conclusions must follow the chosen participant and settlement model.

9 Conclusion

Convex electricity supply need not support an indivisible electric-bus fleet at its own marginal prices, even with continuous charging and full terminal replenishment. The obstruction persists in an explicit hourly generator model with ramps. For nonsmooth supply, existence of support concerns an admissible price set, not an arbitrary solver-selected price. Replication can make the planning gap small without making the incentive of a whole-fleet operator small along every subsequence, while incentives for reserved pairs vanish. Analytical cardinality and energy bounds strengthen the public computations. Joint physical, hull and regret reporting leaves public price support unresolved and separates incumbent incentives from optimal-fleet incentives. Establishing the mechanism’s practical prevalence and an effective computational method requires further independent timetable evidence.

Data and reproducibility

The research repository contains the analytical constructions, declared public-data transformation, experiment protocols, complete outcome tables including unsuccessful attempts, and figure-generation sources. A companion source map links the manuscript’s claims to reviewed evidence and distinguishes exact ideal-model calculations from tolerance-qualified native results. All cases reported here are development cases. No private operational data or reserved evaluation outcomes are used. Public table revisions retain archived results. The new generation example combines exact affine-branch and rational dual certificates with small LP checks; the dual-coordination replay uses closed-form fleet and supply responses. Its full trace distinguishes current from best-so-far dual bounds. No new timetable-scale optimization is claimed. The separate development supplement retains earlier computational comparisons and failures.

A Proofs for the complete-fleet statements

Proof of Proposition 1. For physical x , $p \in \partial F(e(x))$, and any hull point (e, c) , linear minimization over the convex hull gives $c + p^\top e \geq V(p)$. The subgradient inequality gives $F(e) \geq F(e(x)) + p^\top (e - e(x))$. Adding and minimizing over the hull yields $CH \geq c(x) + F(e(x)) - r(x; p)$, proving the first inequality; $c(x) + F(e(x)) \geq D \geq CH$ proves the second. If $r(x; p) = 0$, all inequalities collapse to $CH = D = c(x) + F(e(x))$. Conversely, compactness supplies a physical minimizer

when $D = CH$. Under the relative-interior condition, Fenchel–Rockafellar duality gives an attained maximum $CH = V(p^*) - F^*(p^*)$. Under the alternative minorant condition, apply the same duality theorem to that finite minorant. It agrees with F on all hull loads, while its conjugate bounds F^* from above; weak duality for F then makes the same p^* optimal for F . For any physical minimizer x^* , the two nonnegative Fenchel terms satisfy

$$r(x^*; p^*) + F(e(x^*)) - (p^*)^\top e(x^*) + F^*(p^*) = D - CH = 0.$$

Each term is therefore zero. Fenchel equality gives $p^* \in \partial F(e(x^*))$ and the first equality gives $r(x^*; p^*) = 0$. This holds for every physical minimizer and every attained hull-dual maximizer. $\square$

For the accounting identity, $F^*(p) \geq p^\top e(x) - F(e(x))$ by definition, so supplier LOC is nonnegative. Substitution in (4) gives the claimed sum. The same qualification above yields an attained hull-dual price p^* and $V(p^*) - F^*(p^*) = CH$. At a physical planner optimizer the LOC sum is $D - CH$. At any price in the plan’s subdifferential, Fenchel equality makes supplier LOC zero.

Proof of (6). Let (e^*, c^*) be a hull minimizer, $p^* = a + Qe^*$, $d = \|e(x) - e^*\|_Q$, and $H = H(x)$. First-order hull optimality makes (e^*, c^*) a minimizer of $c + (p^*)^\top e$, so

$$\text{LOC}_{\text{fleet}}(x; p^*) = H - \frac{1}{2}d^2 \geq 0.$$

In particular $d \leq \sqrt{2H}$. For each physical alternative y , the change in its linear advantage when the price moves from p^* to $p_x = p^* + Q(e(x) - e^*)$ is $(e(x) - e^*)^\top Q(e(x) - e(y))$, at most dR_Q by the seminorm Cauchy–Schwarz inequality. Maximizing over y gives $r(x) \leq H - d^2/2 + dR_Q \leq H + R_Q\sqrt{2H}$. The lower inequality is Proposition 1. $\square$

Proof of Proposition 2. Let λ be the one-bus mixture weight and x the mean early load. Complete-plan mixtures have $0 \leq \lambda \leq 1$, $10\lambda \leq x \leq 10$, and intrinsic cost $14 - 7\lambda$. These conditions are sufficient: for $\lambda < 1$ the two-bus component buys $(x - 10\lambda)/(1 - \lambda) \in [0, 10]$ early; at $\lambda = 1$, $x = 10$. For fixed x , cheapest intrinsic cost uses $\lambda = x/10$. Substituting $z = 30 - x$ into (7) gives hull objective

$$14 - \frac{7x}{10} + F(x, 30 - x) = 104 - 2.7x + 0.2x^2 = \frac{7591}{80} + \frac{(x - 27/4)^2}{5}.$$

Its minimizer is $x = 27/4$, $\lambda = 27/40$. Direct minimization of the one- and two-bus physical branches gives respectively 97 and 99. At $(10, 20)$ the gradient of F is $(6, 4)$; its private objective is $7 + 6(10) + 4(20) = 147$, while the two-bus $x = 0$ plan has $14 + 4(30) = 134$. At hull load $(27/4, 93/4)$ the gradient is $(107/20, 93/20) = (5.35, 4.65)$. Both supporting plans have private objective 153.5, and $F^*(p) = 58.6125$ on $\mathbb{R}_+^2$. The hull dual value is therefore 94.8875. The physical one-bus plan is a price minimizer, so fleet LOC is zero; supplier LOC is $97 - 94.8875$. $\square$

Proof of Proposition 3. For reserve $0 \leq r \leq 5$, efficiency η , one-hour early limit P , and one-hour single terminal connector K , write

$$T = 30/\eta, \quad \ell = \max\{0, T - K\}, \quad u = \min\{P, 15/\eta\}, \quad h = \max\{\ell, (10 + r)/\eta\}.$$

The two-bus physical early-energy interval is $[\ell, u]$, and the one-bus interval is $[h, u]$ when nonempty. The lower limit ℓ enforces terminal connector capacity, while $(10 + r)/\eta$ ensures the one bus can begin B above reserve. For every point in these intervals, charge only A’s bus early; terminal demands total $T - x \leq K$ and can be served serially in the one-hour terminal window. Individual inventories and acceptance limits hold, and all used buses finish at capacity.

Write $G_T(x) = F(x, T - x)$, $k = 1/5$, and $m = T/2 - 5a/2$ for early linear coefficient $a = 4$. Then $G_T(x) = G_T(m) + k(x - m)^2$. Suppose $\ell < m < h < u$, $f > 0$, and $f < 2k(h - \ell)(h - m)$. The cheapest hull operating-cost boundary over $[\ell, h]$ is $2f - f(x - \ell)/(h - \ell)$, obtained by mixing the two-bus endpoint ℓ and one-bus endpoint h ; over $[h, u]$ it is f . Put $d = h - \ell$. The strict inequality places the hull minimum $x^* = m + f/(2kd)$ inside (m, h) . Physical branch minimization gives

$$D = G_T(m) + \min\{f + k(h - m)^2, 2f\}, \quad CH = G_T(m) + 2f - \frac{f(m - \ell)}{d} - \frac{f^2}{4kd^2}.$$

Consequently

$$\Delta = \min \left\{ k(h - x^*)^2, \frac{f(m - \ell)}{d} + \frac{f^2}{4kd^2} \right\} > 0.$$

At $(r, \eta, P, K, f, a) = (1, 19/20, 12, 30, 7, 4)$, $(\ell, m, h, u, d) = (30/19, 110/19, 220/19, 12, 10)$. The inequalities are strict: $u - h = 8/19$ and $2kd(h - m) = 440/19 > 7$. Rational substitution yields (9). All strict inequalities persist under small changes in (r, η, P) with $K = 30$ fixed. $\square$

Proof of Proposition 4. For m paired buses, total early energy E ranges over $[10m, 10n]$, with intrinsic cost $14n - 7m$ and terminal energy $30n - E$. Convexifying complete plans gives a triangular projected set; its cheapest operating cost at each E is $14n - 0.7E$. Substitution in (10) gives a quadratic minimized at $E = 27n/4$ with value $7591n/80$.

For fixed integer $m \geq n/2$, physical supply cost is minimized at $E = 10m$, and the total objective is $104n - 27m + 20m^2/n = 7591n/80 + 20(m - 27n/40)^2/n$. For $m \leq n/2$, the best early load is $E = 5n$ when feasible and the branch objective is $99n - 7m$, decreasing toward $n/2$. For odd n , moving from $m = \lfloor n/2 \rfloor$ to $m = \lceil n/2 \rceil$ changes the minimized objective by $-7 + 5/n < 0$. For even n , the two formulas agree at $m = n/2$, and an integer nearest to $27n/40$ lies in the upper regime. Thus the nearest integer to $27n/40$ is optimal, with both integers at a half-integer tie and $|\delta| \leq 1/2$.

At a physical optimum, the price is $p = (4 + 2m/n, 6 - 2m/n)$. The one-bus private objective minus the best two-bus private objective is $40\delta/n$; the latter buys zero early energy when $m/n \geq 1/2$. Hence whole-operator regret is

$$r_n = \begin{cases} m 40\delta/n, & \delta \geq 0, \\ (n - m)(-40\delta/n), & \delta < 0. \end{cases}$$

For $n = 40k + 1$, $m = 27k + 1$ and $\delta = 13/40$, giving (12). The same difference is a reserved pair's individual regret when its current choice is worse; $|40\delta/n| \leq 20/n$. Summing the disadvantaged pairs reproduces the whole-operator expression because each pair has its own reserved connector rights. $\square$

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